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ARTICLE

Nutrition and Health (including climate and ecological aspects)

Skeletal muscle estimation using magnetic-resonance-imaging-based equations for dual-energy X-ray absorptiometry and bioelectrical impedance analysis

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BACKGROUND/OBJECTIVES: Skeletal muscle mass (SMM) estimation is important but challenging in clinical settings. Criterion methods, such as magnetic resonance imaging (MRI), are often inaccessible. However, surrogate methods, such as dual-energy X-ray absorptiometry (DXA) and multi-frequency bioelectrical impedance analysis (MFBIA), can use MRI-based equations to estimate SMM, although the agreement between these methods is unclear.
SUBJECTS/METHODS: Total and segmental SMM were estimated with DXA and MFBIA using MRI-based equations in 313 healthy adults (120 M, 193 F; age 30.2 ± 13.0 y; BMI 24.6 ± 4.0 kg/m^2). DXA total SMM was estimated using the Kim and McCarthy equations, and segmental SMM was estimated using the McCarthy equations. Relationships between DXA and MFBIA SMM were examined using Deming regression, Lin's concordance correlation coefficient (CCC), equivalence testing, Bland-Altman analysis, and related tests.
RESULTS: Strong linear relationships were observed for total (R^2 0.95, CCC 0.96–0.97), leg (R^2 0.90, CCC 0.85) and arm (R^2 0.93, CCC 0.93) SMM in the entire sample. Kim equation SMM demonstrated statistical equivalence with MFBIA for total SMM, but the Deming regression slope differed from 1 and proportional bias was present. McCarthy equation total SMM exhibited a regression slope that did not differ from 1, and no proportional bias was present in the entire sample. However, equivalence with MFBIA was not observed. Systematically higher leg and arm SMM values were observed with DXA as compared to MFBIA.
CONCLUSIONS: While DXA and MFBIA total SMM generally exhibited strong agreement, higher appendicular SMM by DXA highlights technical differences between methods.

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INTRODUCTION

Body composition influences numerous health outcomes, longevity, and disease states, making its assessment advantageous in clinical settings [1, 2]. Skeletal muscle is the largest body compartment in the organ/tissue level of body composition [3, 4], and appropriate skeletal muscle structure and function are critical for performing activities of daily living and maintaining independence with aging [5]. Due to the vital roles of skeletal muscle as the body's protein reserve, agent for locomotion, and site of metabolic processes, the estimation of skeletal muscle mass (SMM) is warranted in health and disease. Indeed, SMM estimation is a primary component of diagnosing and monitoring diseases such as sarcopenia, sarcopenic obesity, frailty, and cachexia [6]. Therefore, methods that accurately quantify SMM are essential as a component of health evaluation and the management of multifarious disease states.

Magnetic resonance imaging (MRI) is considered a criterion method for SMM estimation but is not accessible in many settings. Dual-energy X-ray absorptiometry (DXA) and bioelectrical impedance analysis (BIA) are relatively more affordable and accessible methods that are often used to estimate SMM in research and clinical practice. Data arising from DXA and BIA assessments can be used within prediction equations to estimate whole-body or regional SMM [7–10]. For DXA, the equations of Kim et al. [7, 10], which primarily use appendicular lean mass (ALM) to predict total SMM from MRI, have commonly been used over the past two decades. Recently, McCarthy et al. [9] developed updated equations to predict total and segmental SMM from MRI using total or segmental ALM from DXA. While the software associated with most BIA analyzers may not include validated SMM estimates, select manufacturers have produced such equations for their analyzers. For example, Bosy-Westphal et al. [8] developed and validated SMM prediction equations using both standing and supine phase-sensitive eight-electrode analyzers that allowed for segmental analysis of the body, similar to DXA, resulting in total and segmental SMM estimates.

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Although both DXA and BIA SMM prediction equations developed using MRI as the criterion method are implemented in clinical and research settings, the comparability of these SMM estimates has not yet been established. The shared use of MRI SMM estimates in the development and validation of the equations could promote similarity in the resultant SMM estimates; however, the notable technological differences between DXA – an imaging technique based on attenuation of X-rays – and BIA – reliant on the bioelectrical resistance and reactance of body tissues – could instead result in dissimilarities. Therefore, the purpose of this analysis was to compare total and segmental SMM estimates arising from MRI-based SMM prediction equations for DXA and BIA. It was hypothesized that total SMM would be similar when estimated using DXA and BIA equations, but that segmental SMM estimates would differ due to variability in how segments are detected by DXA versus segmental BIA technology.

MATERIALS AND METHODS

Participants and datasets

Data for the present secondary analysis were pooled from five research studies conducted in the same laboratory from 2018–2021, each recruiting generally healthy adult participants. All studies used the same DXA and multi-frequency BIA (MFBIA) devices within the same research laboratory, with weight and height assessed using a digital scale and mechanical stadiometer (Seca 769, Seca® gmbh & co, Hamburg, Germany). All studies were approved by the Texas Tech University Institutional Review Board (IRB2017-912, IRB2018-417, IRB2019-356, IRB2020-813, and IRB2021-107) [11–15]. Informed consent was obtained from all participants.

Multi-frequency bioelectrical impedance analysis

MFBIA was performed using a 19-frequency, 8-point device (mBCA 515/514, Seca® gmbh & co, Hamburg, Germany) with contact electrodes and assessments conducted in the standing position. This analyzer utilizes frequencies ranging from 1 to 1000 kHz, a measuring current of 100 µA, and has previously been validated against MRI for SMM estimates [8, 16, 17]. Default output of the analyzer includes estimates of total SMM, arm SMM, and leg SMM, using proprietary equations developed in a sample of 136 Caucasian males and females aged 18 to 65 years with BMI ranging from 20.0 to 34.7 kg/m^2 [8]. These equations produced R^2 and RMSE values of 0.97 and 1.2 kg for total SMM, 0.94 and 0.27 kg for arm SMM, and 0.91 and 0.76 kg for leg SMM using the mBCA 515/514 analyzer. In each prediction equation, the primary predictor variable was the impedance index (i.e., height^2/resistance at the 50 kHz frequency [R_{50}] for the corresponding body segment), with reactance at the 50 kHz frequency (Xc_{50}), R_{50}, weight, sex, age and the trunk/extremity indices of R_{50} and Xc_{50} as additional predictors in the models [8].

Using the mBCA 515/514 analyzer in our laboratory, the relative technical error of measurement (TEM) for repeated assessments during one laboratory visit, with repositioning of the participant between assessments, is 0.4% for total SMM, 0.4% for arm SMM, and 0.3% for leg SMM. The TEM for assessments on consecutive days is 1.6% for total SMM, 2.7% for arm SMM, and 1.6% for leg SMM.

Dual-energy X-ray absorptiometry

DXA scans were performed on a Lunar Prodigy scanner (General Electric, Boston, MA, USA) with enCORE software (v. 16.2). The scanner was calibrated daily prior to scanning using a manufacturer-supplied calibration block. Positioning of participants took place using custom-made foam blocks to promote consistency of positioning [18, 19]. Foam blocks were placed bilaterally between the hands and hips, with the hands placed in a neutral position. A block and strap at the feet ensured consistent foot positioning and orientation of the feet perpendicular to the scanning table. In the event that a participant was too broad to fit within DXA's scanning dimensions, excluded body portions were estimated via reflection scanning techniques, which introduce minimal error [20, 21]. A trained operator manually adjusted region of interest lines within the enCORE software to specify body segments (i.e., arms, trunk, and legs). DXA estimates of tissue and region BF% were obtained.

DXA appendicular lean mass (ALM), calculated as the sum of lean soft tissue in both arms and both legs, was used to estimate total SMM using two published equations validated against MRI: Kim et al. 2004 [7] and McCarthy et al. 2023 [9]. Additionally, SMM of the arms and legs were estimated from DXA arm and leg lean soft tissue, respectively, using the equations of McCarthy et al. [9].

Kim et al. 2004 [7]:

$$SMM_{total} = 1.19^{*}ALM - 1.65$$

McCarthy et al. 2023 [9]:

$$SMM_{total} = 1.12^{*}ALM - 0.63$$

$$SMM_{arms} = 0.58^{*}LMarms + 0.15$$

$$SMM_{legs} = 0.78^{*}LMlegs - 1.07$$

Using the Lunar Prodigy scanner in our laboratory, the TEM for repeated assessments during one laboratory visit, with repositioning of the participant between assessments, is 1.4% for total SMM, 2.0% for arm SMM, and 1.6% for leg SMM, using the equations of McCarthy et al. [9]. The TEM for assessments on consecutive days is 1.3% for total SMM, 4.3% for arm SMM, and 1.0% for leg SMM. Using the Kim et al. [7]. total SMM equation, the TEM for repeated assessments during one laboratory visit, with repositioning of the participant between assessments, is 1.4% for total SMM. The TEM for assessments on consecutive days is 1.3% for total SMM.

Statistical analysis

The linear relationships between SMM_{DXA} and SMM_{MFBIA} were established using Deming regression, which accounts for errors in the measurement of both variables [22], and compared to a perfect relationship (i.e., the line of identity). Pearson's R^2, root mean square error (RMSE), and Lin's concordance correlation coefficient (CCC) values were also calculated. Equivalence testing [23] was performed using two one-sided *t*-tests (TOST) to determine if SMM_{DXA} and SMM_{MFBIA} were equivalent based on equivalence regions of 2.5% for total SMM and 5% for arm and leg SMM, consistent with previous research [9], alongside paired-samples *t*-tests. Additionally, Bland-Altman analysis was performed [24], including estimation of the 95% limits of agreement and linear regression to examine proportional bias. For completeness, analyses were performed in the entire sample ($n = 313$), males only ($n = 120$), and females only ($n = 193$). Statistical analyses were conducted in R (version 4.2.1) using the *DescTools* [25], *deming* [22], and *TOSTER* [23] packages. Values are presented as mean ± SD and statistical significance was accepted at $p < 0.05$.

RESULTS

Participants

Three-hundred and thirteen participants (120 M, 193 F) were included in the present analysis (Table 1). In general, participants were young (30.2 ± 13.0 y) and normal weight or overweight (BMI 24.6 ± 4.0 kg/m^2), although BMI ranged from 17.7 to 42.1 kg/m^2 in males and from 16.1 to 37.7 kg/m^2 in females. Individuals with BMI values ≥ 30.0 accounted for 9.6% of the dataset ($n = 30$). Relative body fat values ranged from 5.6 to 42.6% in males (mean ± SD 20.1 ± 7.7%) and from 12.5 to 50.1% in females (mean ± SD 29.7 ± 7.6%). In the sample, 221 individuals were non-Hispanic Caucasian, 45 were Hispanic Caucasian, 20 were Black, 19 were Asian, 7 were South or Central American, and 1 was a Pacific Islander.

Total skeletal muscle

Using the Kim et al. 2004 equation [7], strong linear relationships were observed between DXA and MFBIA SMM estimates in the entire sample and each sex individually (R^2: 0.83 to 0.95; $p < 0.001$ for each), with RMSE values of 1.54 to 1.74 kg (Fig. 1). CCC values were 0.97, 0.93, and 0.90 in the entire sample, males, and females, respectively. The slope of the Deming regression line significantly differed from the line of identity in the entire sample (95% CI: 0.93, 0.99) and males only (95% CI: 0.76, 0.86), but not females only (95% CI: 0.83, 1.00). Based on the results of equivalence testing,

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Table 1. Participant Characteristics.

	All ($n = 313$)[a]	Males ($n = 120$)	Females ($n = 193$)
Age (y)	30.2 ± 13.0	31.5 ± 13.1	29.4 ± 12.9
Height (cm)	169.7 ± 9.1	177.1 ± 7.4	165.0 ± 6.6
Weight (kg)	71.3 ± 14.7	81.5 ± 13.3	64.9 ± 11.7
BMI (kg/m^2)	24.6 ± 4.0	26.0 ± 4.0	23.8 ± 3.8
FFMI (kg/m^2)[b]	18.0 ± 2.6	20.5 ± 2.1	16.5 + 1.5
FMI (kg/m^2)[b]	6.6 ± 3.1	5.4 ± 2.9	7.3 ± 3.0
Body fat (%)[b]	26.0 ± 8.9	20.1 ± 7.7	29.7 ± 7.6
SMMtotal, DXA - Kim equation (kg)[c]	24.8 ± 6.9*	31.6 ± 5.2*	20.6 ± 3.6*
SMMtotal, DXA - McCarthy equation (kg)[c]	24.3 ± 6.5*	30.7 ± 4.9*	20.3 ± 3.3*
SMMtotal, MFBIA (kg)	25.3 ± 6.5	32.2 ± 4.4	21.0 ± 3.2
SMMlegs, DXA (kg)[c]	11.8 ± 3.0*	14.6 ± 2.4*	10.1 ± 1.8*
SMMlegs, MFBIA (kg)	10.7 ± 2.3	12.9 ± 1.8	9.4 ± 1.5
SMMarms, DXA (kg)[c]	3.5 ± 1.2*	4.7 ± 0.9*	2.7 ± 0.5*
SMMarms, MFBIA (kg)	3.2 ± 1.0	4.3 ± 0.8	2.5 ± 0.5

BMI body mass index, *DXA* dual-energy X-ray absorptiometry, *FFMI* fat-free mass index, *FMI* fat mass index, *MFBIA* multi-frequency bioelectrical impedance analysis, *SMM* skeletal muscle mass.
*A significant difference from the corresponding MFBIA variable ($p \leq 0.001$), based on paired-samples *t*-tests.
[a]Values displayed as mean ± SD.
[b]FFMI, FMI, and body fat values taken from MFBIA (Seca mBCA 515/514).
[c]For each DXA SMM variable.

total SMM estimated by DXA and MFBIA was statistically equivalent in the entire sample (TOST interval: 0.32, 0.61 kg; $p = 0.04$) and males only (TOST interval: 0.26, 0.77 kg; $p = 0.03$), but not females only (TOST interval: 0.26, 0.62 kg; $p = 0.22$). From Bland-Altman analysis, proportional bias was observed in the entire sample (95% CI for slope: −0.07, −0.02), males only (95% CI for slope: −0.24, −0.13), and females only (95% CI for slope: −0.19, −0.06).

Using the McCarthy et al. 2023 equation [9], strong linear relationships were observed between DXA and MFBIA SMM estimates in the entire sample and each sex individually (R^2: 0.83 to 0.95; $p < 0.001$ for each), with RMSE values of 1.57 to 2.08 kg (Fig. 2). CCC values were 0.96, 0.90, and 0.89 in the entire sample, males, and females, respectively. The slope of the Deming regression line did not differ from the line of identity in the entire sample (95% CI: 0.99, 1.05) or females only (95% CI: 0.88, 1.08), but differed in males only (95% CI: 0.81, 0.92). Based on the results of equivalence testing, total SMM estimated by DXA and MFBIA was not statistically equivalent in the entire sample (TOST interval: 0.87, 1.14 kg; $p = 1.0$), males only (TOST interval: 1.23, 1.68 kg; $p = 1.0$), or females only (TOST interval: 0.56, 0.89 kg; $p = 1.0$). From Bland-Altman analysis, proportional bias was not present in the entire sample (95% CI for slope: −0.01, 0.04), but was present in males only (95% CI for slope: −0.18, −0.06), and females only (95% CI for slope: −0.13, −0.00).

Leg skeletal muscle

Moderate to strong linear relationships were observed between DXA and MFBIA leg SMM estimates in the entire sample and each sex individually (R^2: 0.76 to 0.90; $p < 0.001$ for each), with RMSE values of 1.12 to 2.0 kg (Fig. 3). CCC values were 0.85, 0.66, and 0.78 in the entire sample, males, and females, respectively. The slope of the Deming regression line differed from the line of identity in the entire sample (95% CI: 0.74, 0.82), males only (95% CI: 0.64, 0.76), and females only (95% CI: 0.73, 0.92). Based on the results of equivalence testing, leg SMM estimated by DXA and MFBIA was statistically equivalent in the entire sample (TOST interval: −1.18, −0.98 kg; $p = 0.004$) and females only (TOST interval: −0.79, −0.58 kg; $p < 0.001$), but not males only (TOST interval: −1.87, −1.54 kg; $p = 0.92$). From Bland-Altman analysis, proportional bias was present in the entire sample (95% CI for slope: −0.22, −0.29), males only (95% CI for slope: −0.23, −0.38), and females only (95% CI for slope: −0.15, −0.30).

Arm skeletal muscle

Moderate to strong linear relationships were observed between DXA and MFBIA arm SMM estimates in the entire sample and each sex individually (R^2: 0.72 to 0.93; $p < 0.001$ for each), with RMSE values of 0.29 to 0.57 kg (Fig. 4). CCC values were 0.93, 0.79, and 0.80 in the entire sample, males, and females, respectively. The slope of the Deming regression line differed from the line of identity in the entire sample (95% CI: 0.87, 0.96) and males only (95% CI: 0.84, 0.98), but not females only (95% CI: 0.97, 1.24). Based on the results of equivalence testing, arm SMM estimated by DXA and MFBIA was statistically equivalent in the entire sample (TOST interval: −0.30, −0.24 kg; $p < 0.001$), males only (TOST interval: −0.51, −0.41 kg; $p < 0.001$) and females only (TOST interval: −0.19, −0.13 kg; $p < 0.001$). From Bland-Altman analysis, proportional bias was present in the entire sample (95% CI for slope: −0.16, −0.10) and males only (95% CI for slope: −0.18, −0.03), but not females only (95% CI for slope: −0.09, 0.07).

DISCUSSION

Skeletal muscle estimation is warranted in numerous clinical settings, although a relative inaccessibility of criterion assessments, such as MRI, often necessitates the use of surrogate methods. In the present analysis, two such techniques, DXA and MFBIA, were investigated to establish their relative agreement when using MRI-based estimation equations to predict total and segmental SMM. An understanding of the comparability of DXA and MFBIA in this context may help inform clinicians and researchers using and interpreting skeletal muscle estimates from these technologies.

For total SMM, the longstanding Kim et al. [7] and newly developed McCarthy et al. [9] DXA equations, both of which estimate SMM without contributions of adipose tissue, demonstrated strong linear relationships with the Bosy-Westphal et al. [8]

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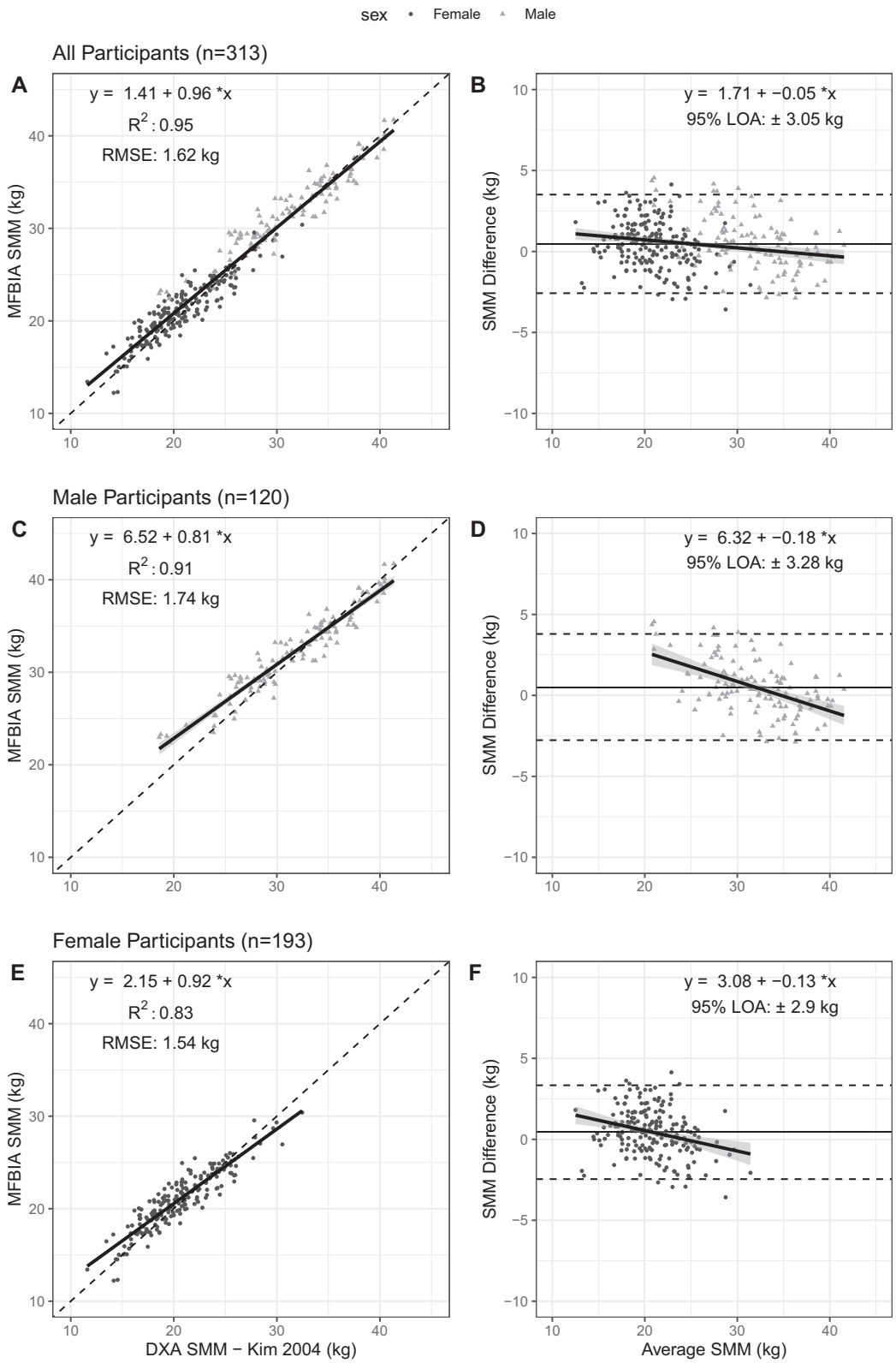

Fig. 1 Total skeletal muscle mass estimated by dual-energy X-ray absorptiometry (Kim et al. 2004 equation) and multi-frequency bioelectrical impedance analysis. The linear relationship between DXA and MFBIA total SMM estimates, as established by Deming regression, is displayed for all participants (**A**), male participants only (**C**), and female participants only (**E**). Bland-Altman analysis is displayed for all participants (**B**), male participants only (**D**), and female participants only (**F**). DXA dual-energy X-ray absorptiometry, LOA limits of agreement, MFBIA multi-frequency bioelectrical impedance analysis, RMSE root mean square error.

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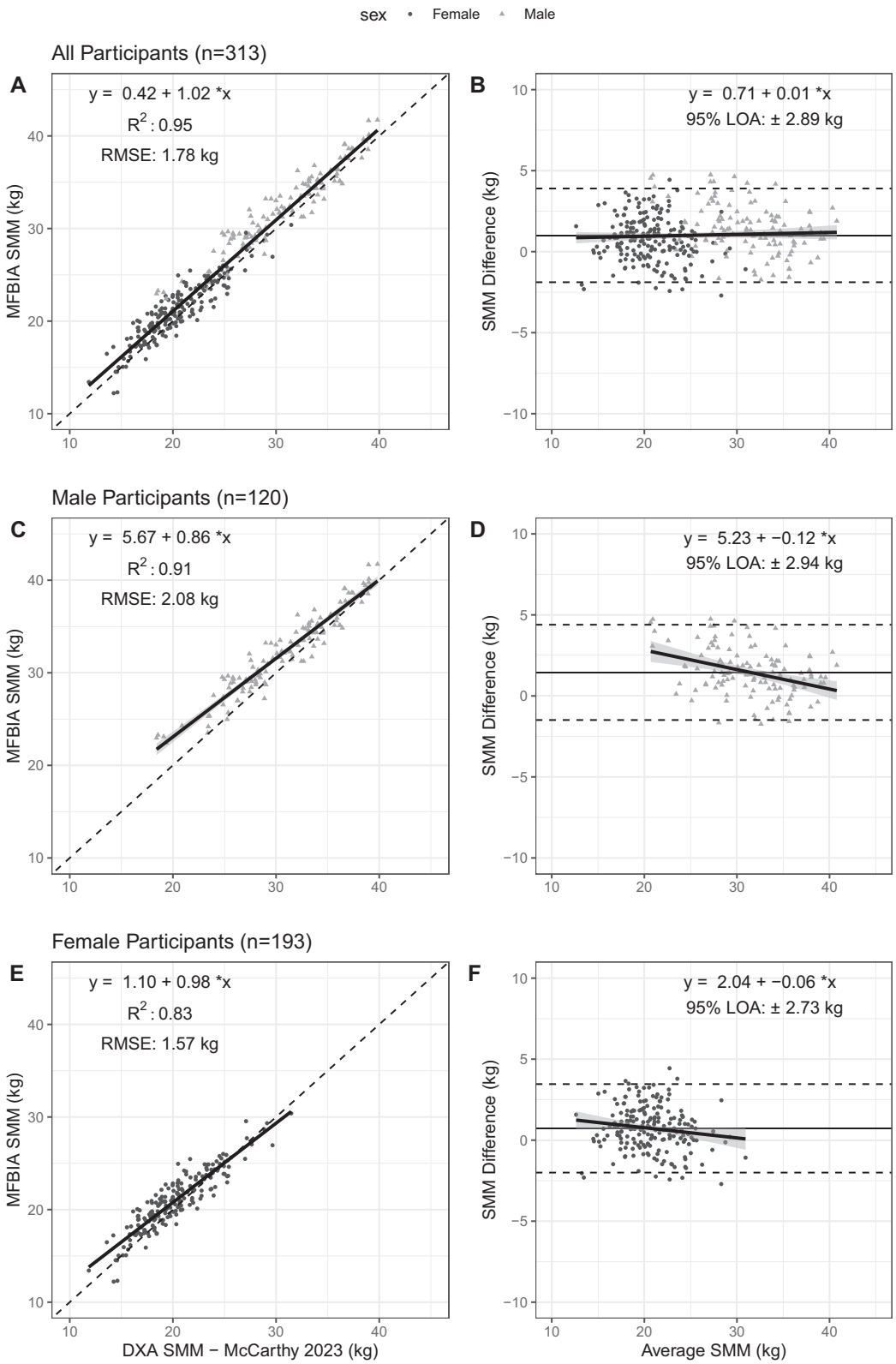

Fig. 2 Total skeletal muscle mass estimated by dual-energy X-ray absorptiometry (McCarthy et al. 2023 equation) and multi-frequency bioelectrical impedance analysis. The linear relationship between DXA and MFBIA total SMM estimates, as established by Deming regression, is displayed for all participants (**A**), male participants only (**C**), and female participants only (**E**). Bland-Altman analysis is displayed for all participants (**B**), male participants only (**D**), and female participants only (**F**). DXA dual-energy X-ray absorptiometry, LOA limits of agreement, MFBIA multi-frequency bioelectrical impedance analysis, RMSE root mean square error.

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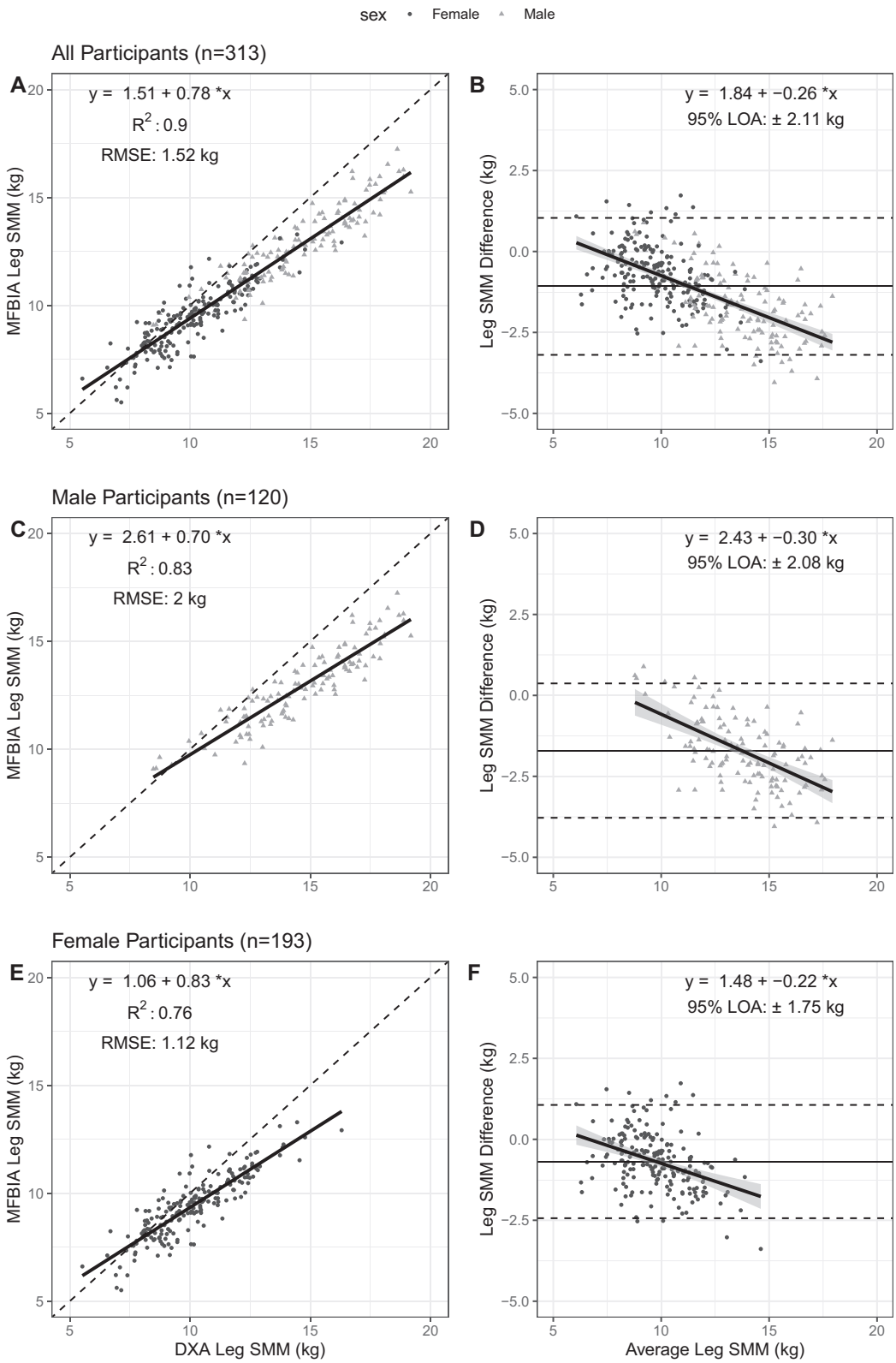

Fig. 3 Leg skeletal muscle mass estimated by dual-energy X-ray absorptiometry and multi-frequency bioelectrical impedance analysis. DXA leg skeletal muscle was estimated using the McCarthy et al. 2023 equation. Values represent the masses of skeletal muscle in the left and right legs combined. The linear relationship between DXA and MFBIA total SMM estimates, as established by Deming regression, is displayed for all participants (**A**), male participants only (**C**), and female participants only (**E**). Bland-Altman analysis is displayed for all participants (**B**), male participants only (**D**), and female participants only (**F**). DXA dual-energy X-ray absorptiometry, LOA limits of agreement, MFBIA multi-frequency bioelectrical impedance analysis, RMSE root mean square error.

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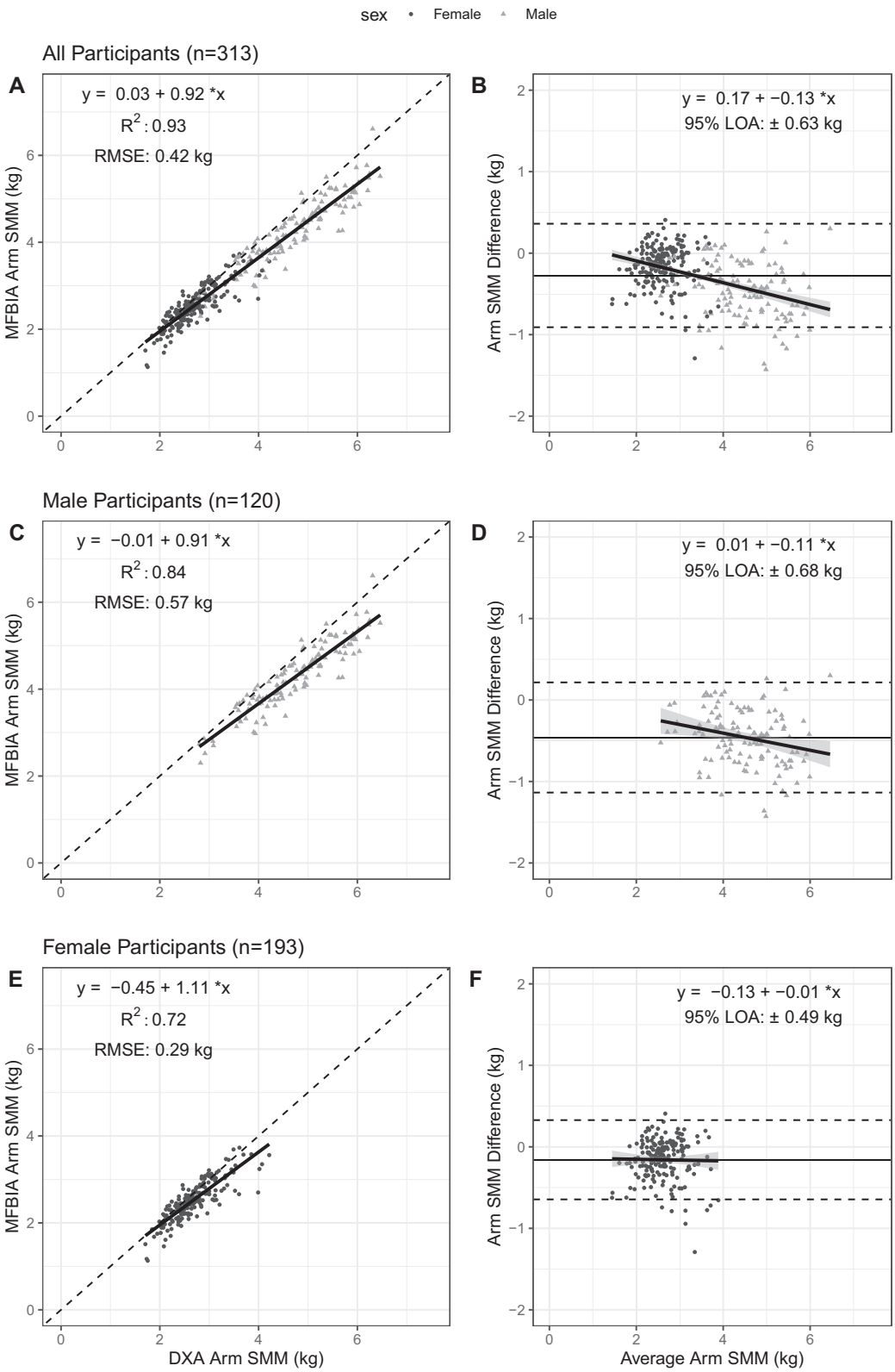

Fig. 4 Arm skeletal muscle mass estimated by dual-energy X-ray absorptiometry and multi-frequency bioelectrical impedance analysis. DXA arm skeletal muscle was estimated using the McCarthy et al. 2023 equation. Values represent the masses of skeletal muscle in the left and right arms combined. The linear relationship between DXA and MFBIA total SMM estimates, as established by Deming regression, is displayed for all participants (**A**), male participants only (**C**), and female participants only (**E**). Bland-Altman analysis is displayed for all participants (**B**), male participants only (**D**), and female participants only (**F**). DXA dual-energy X-ray absorptiometry, LOA limits of agreement, MFBIA multi-frequency bioelectrical impedance analysis, RMSE root mean square error.

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MFBIA SMM estimates. However, although similar RMSE (~1.5 to 2.0 kg) and CCC (~0.90 to 0.97) values were observed with both DXA equations, other aspects of performance differed. Total SMM estimated by the Kim equation exhibited statistical equivalence with MFBIA SMM in the entire samples and males only, but the slope of the Deming regression line characterizing the linear relationship between methods differed from 1 in the same comparisons. Additionally, Bland-Altman analysis revealed proportional bias (i.e., differential agreement based on the quantity of SMM) in the entire sample, males, and females. In contrast, the slope of the Deming regression line for DXA SMM estimated using the McCarthy equation and MFBIA differed from 1 in males only, but not in the entire sample or females only. Additionally, proportional bias was not observed for the McCarthy equation in the entire sample but was present when sexes were considered individually. The systematic difference between MFBIA and DXA SMM when using the McCarthy equation was ~1 kg on average, as compared to ~0.5 kg on average with the Kim equation, such that statistical equivalence was not demonstrated in any analysis using total SMM from the McCarthy equation. Collectively, these results suggest a generally strong level of agreement between DXA and MFBIA total SMM estimates, with differences in select performance metrics based on which DXA SMM estimation equation is employed.

A unique aspect of DXA and the MFBIA analyzer employed in the present analysis is their ability to provide segmental body composition estimates, although the underlying methods of separating specific body regions differs [19, 26]. While DXA scan processing includes automated or manual specification of regions of interest to categorize tissue, MFBIA employs algorithms based on differences in bioelectrical properties of larger diameter (i.e., trunk) and smaller diameter (i.e., appendages) regions of the body [27]. These technological differences are expected to contribute to poorer agreement between segmental SMM estimates as compared to total SMM. For example, discrepancies in leg SMM could be due to some SMM in the pelvic region, such as portions of the gluteal muscles, being considered part of the leg based on standard DXA region of interest lines, whereas this may not be the case for impedance-based technologies. This contention is supported by the present data, which demonstrated systematically higher leg SMM estimated by DXA as compared to MFBIA (DXA 11.8 ± 3.8 kg, MFBIA 10.7 ± 2.3 kg; $p < 0.001$ via *t*-test). Furthermore, the discrepancy in leg SMM between methods grew larger as leg SMM increased. Interestingly, although the anatomical transition from the trunk to the arm could be expected to be more similar between DXA and MFBIA, a similar finding of systematically higher arm SMM by DXA was observed (DXA 3.5 ± 1.2 kg, MFBIA 3.2 ± 1.0 kg; $p < 0.001$), although the magnitude of proportional bias was smaller. The higher appendicular SMM estimated by DXA was observed alongside higher total SMM from MFBIA (DXA 24.3 ± 6.5 kg via McCarthy equation, MFBIA 25.3 ± 6.5 kg; $p < 0.001$). This indicates that, although trunk SMM estimation equations were not directly employed, the SMM of the trunk could be expected to be ~2.5 kg higher on average when estimated by MFBIA as compared to DXA. Importantly, DXA-based SMM estimation typically uses ALM, rather than total lean mass, due to the confounding influence of non-skeletal-muscle lean mass in the trunk region, notably the viscera [7, 9, 10].

Several important considerations should be noted. First, the purpose of this analysis was to establish the agreement between two common surrogate methods of SMM estimation. Both DXA and MFBIA SMM estimation methods in the present analysis were based on MRI, and neither can be considered a criterion relative to the other. Nonetheless, establishing similarities and dissimilarities of total and segmental SMM estimated by these methods can help clinicians and researchers interpret data obtained across devices. Second, the participants used in the development of each equation should be considered, along with the characteristics of the present sample. The Kim et al. [7] equation was developed in healthy, non-exercising, racially diverse adults (n 170 [96 M, 174 F], age ~46 ± 18 y, BMI ~ 25 ± 4 kg/m^2), while the equations of McCarthy et al. [9] were developed in healthy Caucasian adults engaged in regular physical activity but not programmed exercise training (n 380 [168 M, 212 F], age ~50 ± 18 y, BMI ~ 26 ± 4 kg/m^2). The MFBIA SMM equations of Bosy-Westphal et al. [8] were developed in a sample of Caucasian adults (n 136 [70 M, 66 F], age ~40 ± 12 y, BMI ~ 25 ± 4 kg/m^2), although subsequent validation took place in a racially diverse sample. In the present analysis, participants were younger than the development samples (~30 ± 13 y) although BMI values were similar (~25 ± 4 kg/m^2). Although 70% of the sample self-reported being non-Hispanic Caucasian, the sample also included Hispanic Caucasian (14%), Black (6%), and Asian (6%) individuals, among others (~3%). Many of the participants in this sample were exercise-trained, which is dissimilar to several of the validation samples. However, total SMM was similar, with values of ~31 to 32 and ~20 to 21 kg in males and females, respectively, in the present study, as compared to ~32 and ~20 kg [7] and ~31 and ~20 kg [9] in the previous studies. While not presented by sex, total SMM in the entire sample of Bosy-Westphal et al. [8] (~27 ± 7 kg) was slightly higher than the present sample (~25 ± 7 kg), which may be due to the higher proportion of males in the former investigation (51% vs. 38% in the present study). The young, generally healthy sample with fairly limited representation of individuals with obesity (~10% of sample based on BMI) should be considered when applying the findings of the present study. Third, the analyzer-specific performance of BIA devices should be noted. The present analyzer exhibits specific hardware and software features that are not shared by all devices. Additionally, it has been subject to more rigorous development and validation procedures than many commercially available products [8, 16, 17]. As such, the present results do not represent the performance of the full spectrum of BIA analyzers used in research and clinical practice. Finally, based on the cross-sectional nature of the analysis, our data do not indicate the ability of the studied methods to assess changes in SMM over time. As gain and loss of SMM are important outcomes in the context of health and disease, future work should establish the relative agreement of these methods for detecting changes in SMM.

In summary, the present investigation demonstrates the level of agreement that may be expected for total and segmental SMM estimates using specific MRI-based estimation equations for DXA and MFBIA. While total SMM generally exhibited strong agreement between methods, higher arm and leg SMM estimates by DXA highlight technical differences between these assessment techniques.

DATA AVAILABILITY

Data may be available from corresponding author, upon reasonable request and pending relevant institutional approval.

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AUTHOR CONTRIBUTIONS

GMT and CL conceived the work. GMT, CR, and MRS contributed to data collection. GMT performed the statistical analysis. GMT and CL wrote the initial draft of the manuscript. All authors contributed to revision of the manuscript, approved the final version, and agree to be accountable for the work.

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ETHICAL APPROVAL

This research was performed in accordance with the Declaration of Helsinki and was approved by the Texas Tech University Institutional Review Board (IRB2017-912, IRB2018-417, IRB2019-356, IRB2020-813, and IRB2021-107).

INFORMED CONSENT

Informed consent was obtained from all participants.

ADDITIONAL INFORMATION

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